Mend-Amar Badral

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Computer Science M.Sc. student with a background in Applied Mathematics and a strong interest in reliable software systems. Experienced in building automation scripts, working in Linux environments, and integrating simulation and data processing pipelines. I value disciplined work habits, steady technical improvement, and collaborative approach to problem solving.

Education

Budapest University of Technology and Economics (BME) — Sep, 2025 – June, 2027
M.S. in Computer Science Engineering — Budapest, Hungary
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) — Sep, 2021 – June, 2025
B.S. in Applied Mathematics — Ulaanbaatar, Mongolia
GPA: 3.2/4.0
Thesis: Evaluating land degradation using image processing methods
Presentation

Experience

Center of Mathematics Application Research Lab — Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- **Lab Leadership:** Selected as Undergraduate Lab Lead in final year. Participated in the weekly research progress meetings.
- **Research assistance:** Enabled reliable vegetation index calculation from large amount of drone imagery by developing a custom, vendor specific **Python program** which takes camera and sensor parameters into account.
- **Organization:** Reduced the reliance on proprietary drone image processing software by switching to open source Docker containerized application. Authored a code and process documentation which made it easier for new students to onboard.

Projects

aiSim Data Integration for 3D Gaussian Splatting

- Worked with aiSim simulation software during Project Laboratory course at BME.
- Implemented a custom JSON parser to extract structured image and sensor data from simulation outputs.
- Integrated parsed data into nerfstudio pipeline to train 3D Gaussian Splat representations.

Skills

Programming:	Python, Julia, C/C++, R, JavaScript
Tools:	Docker, Linux, Bash, Git, GitHub Actions, SSH, LaTeX